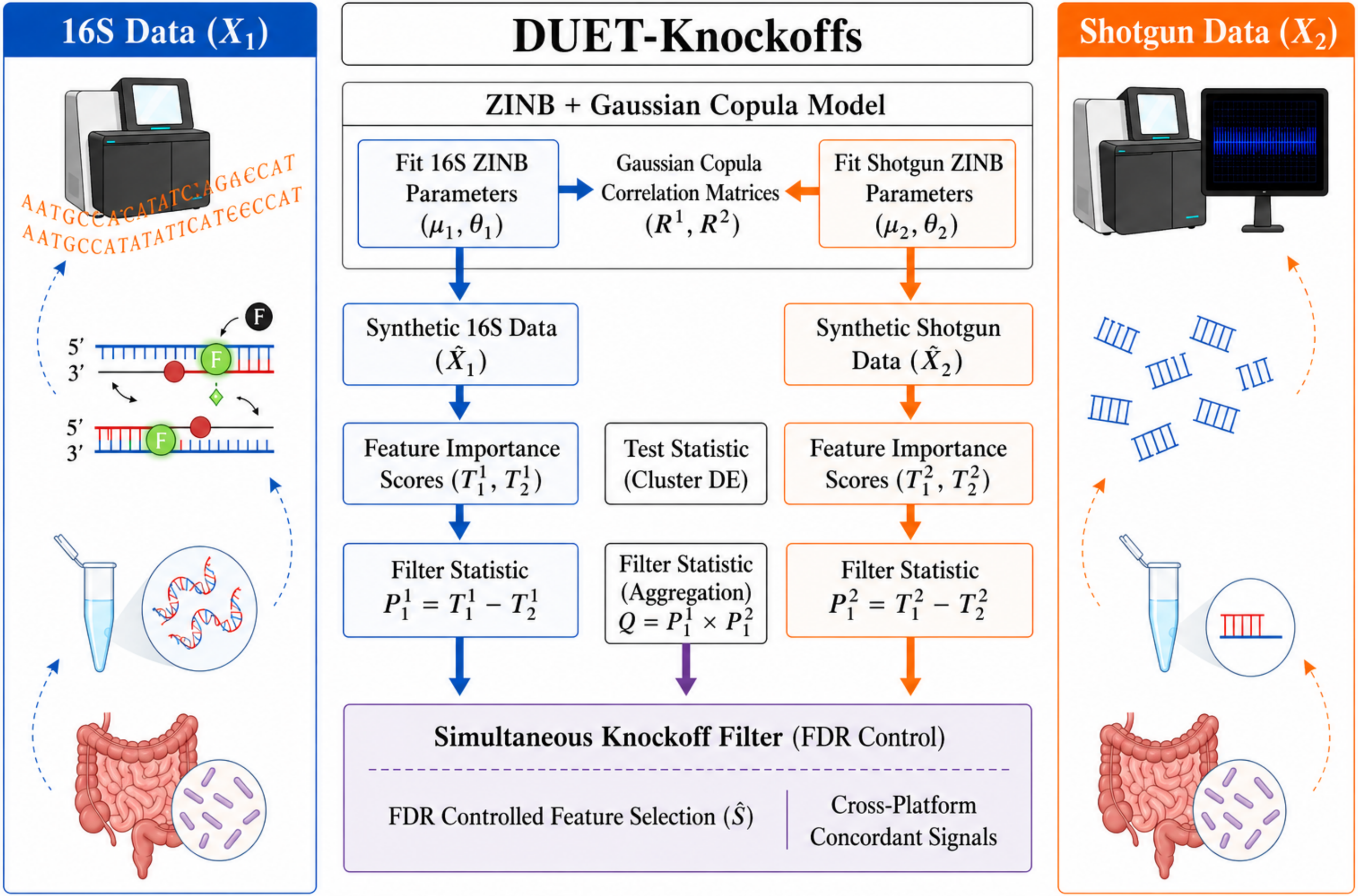
16S Data (X_1)
DUET-Knockoffs
Shotgun Data (X_2)
ZINB + Gaussian Copula Model
Fit 16S ZINB Parameters (μ_1, θ_1)
Gaussian Copula Correlation Matrices (R^1, R^2)
Fit Shotgun ZINB Parameters (μ_2, θ_2)
Synthetic 16S Data ($\tilde{X}_1$)
Synthetic Shotgun Data ($\tilde{X}_2$)
Feature Importance Scores (T_1^1, T_2^1)
Test Statistic (Cluster DE)
Feature Importance Scores (T_1^2, T_2^2)
Filter Statistic $P_1^1 = T_1^1 - T_2^1$
Filter Statistic (Aggregation) $Q = P_1^1 \times P_1^2$
Filter Statistic $P_1^2 = T_1^2 - T_2^2$
Simultaneous Knockoff Filter (FDR Control)
FDR Controlled Feature Selection ($\hat{S}$)
Cross-Platform Concordant Signals
AATGCCATATATCTAGAECAT
AATGCCATATATTCATEECCAT
5'
3'
F